FLOW CHART HOW SYSTEM WORKS

- Deployment: Smart Posts installed at 2km–5km intervals.
- User Scan: Hiker scans QR to open the Offline Portal.
- Data Processing: ESP32 collects environmental and audio data.
- Relay: Data "hops" via LoRa mesh to reach the base.
- Action: Wildlife rangers receive live alerts for rescue.

THE SUMMARY

PathFinder is an Industrial Grade AIoT safety network for remote environments. By bridging the gap between disconnected hiking trails and emergency responders, we ensure that safety is never out of reach, even when signals are.

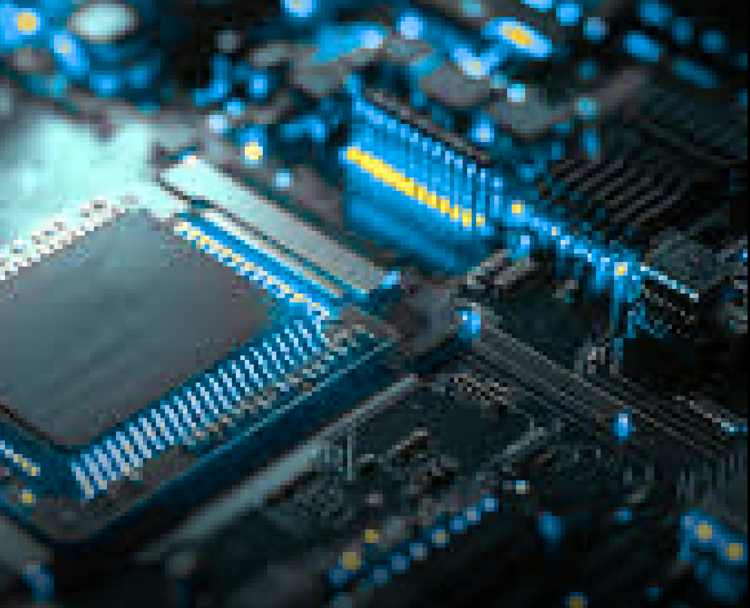
1. Zero Signal Connectivity
2. Human Centric Safety
3. Industry 5.0 Ready

Team GTricks | University of Moratuwa

AIOT FOR A SUSTAINABLE FUTURE
SLIOT CHALLENGE 2026

GTricks PATHFINDER

Smart Posts for
Remote Hiking Trails



TECHNOLOGY

INDUSTRIAL PCB DESIGN

- Engineered with 0.6mm copper tracks and 1.0mm safety clearances to survive moisture, heat, and vibrations in the wild.

LORA COMMUNICATION

- Utilizing the Ra-02 433MHz module for long-range, low-power data transmission over vast geographical areas.

SOLAR POWER MANAGEMENT

- Fully self-sustaining system with integrated solar charging and high-capacity battery for 24/7 operation.

USER INTERFACE

DYNAMIC OFFLINE MAP

- GPS location without internet data.

LIVE ENVIRONMENTAL DATA

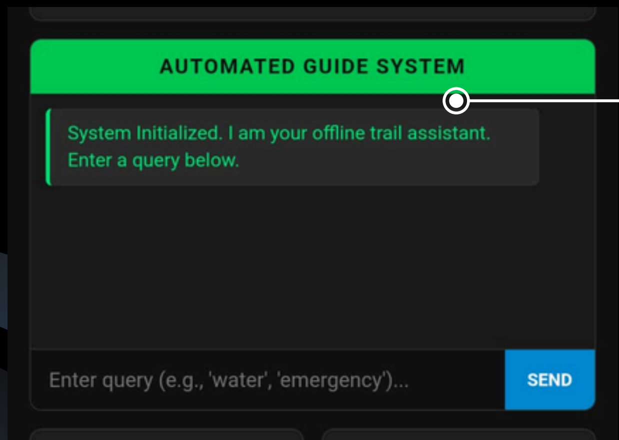
- BMP280 data for flash-flood and storm alerts.

SAFETY STATUS

- Real-time "Safe" or "Danger" indicators.

ONE-TOUCH SOS

- Digital (phone) and Physical (box) buttons.



AI ML PART

- On Device Intelligence: We utilize TinyML to process audio data directly on the ESP32 using the INMP441 MEMS microphone.
- Animal Detection: The model is trained to recognize the specific acoustic frequency of wild animals (like elephants) and heavy vibrations.
- Real Time Alerts: By processing data at the "Edge," the system identifies threats instantly without needing a cloud connection, saving critical seconds in a dangerous situation.